

# ME 2801, HW2, Inverse Laplace Transform

Partial fraction expansion with complex roots.

$$F(s) = \frac{s+5}{(s+2)(s^2+36)}$$

Using the handout on Partial Fraction Expansion with Complex Roots

```
num = [1 5];
den = conv([1 2],[1 0 36])
```

```
den = 1×4
      1      2      36      72
```

```
[R1,P1,K1] = residue(num,den)
```

```
R1 = 3×1 complex
    -0.0375      -      0.0958i
    -0.0375      +      0.0958i
     0.0750      +      0.0000i
P1 = 3×1 complex
     0.0000      +      6.0000i
     0.0000      -      6.0000i
    -2.0000      +      0.0000i
K1 =
[]
```

$$a = 0, \omega = 6, \alpha = -0.0375, \beta = 0.0958$$

$$F(s) = \frac{2(-0.0375)s}{s^2 + 6^2} + \frac{2(0.0958)(6)}{s^2 + 6^2} + \frac{0.075}{s+2}$$

$$f(t) = 2[-0.0375\cos(6t) + 0.0958\sin(6t)] + 0.075e^{-2t}$$

## G(s)

$$G(s) = \frac{2}{s(s^2 + s + 16.25)}$$

```
num = 2;
den = [1 1 16.25 0];
[R2,P2,K2] = residue(num,den)
```

```
R2 = 3×1 complex
    -0.0615      +      0.0077i
    -0.0615      -      0.0077i
     0.1231      +      0.0000i
P2 = 3×1 complex
    -0.5000      +      4.0000i
    -0.5000      -      4.0000i
     0.0000      +      0.0000i
K2 =
[]
```

$$a = 0.5, \omega = 4, \alpha = -0.0615, \beta = -0.0077$$

$$G(s) = 2 \left[ \frac{-0.0615(s + 0.5) - 0.0077(4)}{(s + 0.5)^2 + 4^2} \right] + \frac{0.1231}{s}$$

$$g(t) = 2e^{-0.5t}[-0.0615\cos(4t) - 0.0077\sin(4t)] + 0.1231$$

## H(s)

$$H(s) = \frac{1}{(s^2 + 9)(s^2 + 4s + 1)}$$

```
num = 1;
den = conv([1 0 9],[1 4 1]);
[R3,P3,K3] = residue(num,den)
```

```
R3 = 4×1 complex
   -0.0126      + 0.0000i
   -0.0096      + 0.0064i
   -0.0096      - 0.0064i
    0.0318      + 0.0000i
P3 = 4×1 complex
   -3.7321      + 0.0000i
    0.0000      + 3.0000i
    0.0000      - 3.0000i
   -0.2679      + 0.0000i
K3 =
```

```
[]
```

$$a = 0, \omega = 3, \alpha = -0.0096, \beta = -0.0064$$

$$H(s) = 2 \left[ \frac{-0.0096(s) - 0.0064(3)}{s^2 + 3^2} \right] + \frac{-0.0126}{s + 3.7321} + \frac{0.0318}{s + 0.2679}$$

$$g(t) = 2[-0.096\cos(3t) - 0.0064\sin(3t)] - 0.0126e^{-3.7321t} + 0.0318e^{-0.2679t}$$